



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

SECP3744 – SYSTEM DEVELOPMENT TECHNOLOGY (WBL)
(Student Management System)

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Section 01

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1.0 System Documentation

1.1 System introduction

The Student Management System (SMS) is a comprehensive web-based application designed to facilitate the academic administration of courses and student records. Developed for the SECP3723 System Development Technology course, this project utilizes a traditional Client-Server architecture powered by Vanilla PHP and a MySQL database to ensure data integrity and accessibility. The primary objective of the system is to modernize the course registration process, allowing for real-time management of academic data while replacing inefficient manual methods with a secure, automated digital solution.

The system is architected to serve three distinct user groups: Students, Lecturers, and IT Staff. Students are provided with self-service tools to search for available courses, register for classes, view their academic slips, and manage their registration status through add/drop registration functions. Lecturers are granted access to view their assigned teaching schedules and access detailed lists of enrolled students for their specific sections. IT Staff (Administrators) possess elevated privileges to manage the system's core data, including the ability to add, modify, or delete course details and oversee the dashboard statistics.

To ensure robustness and security, the application implements critical functions such as password hashing and SQL injection prevention, adhering to modern development standards. A key feature of the system is the auto-approval logic for registrations, which streamlines the enrollment process by automatically validating class capacity limits before confirming a student's placement. This comprehensive approach ensures a user-friendly experience while maintaining strict data validation and security protocols across all modules.

1.2 System Design (Use Case, ERD)

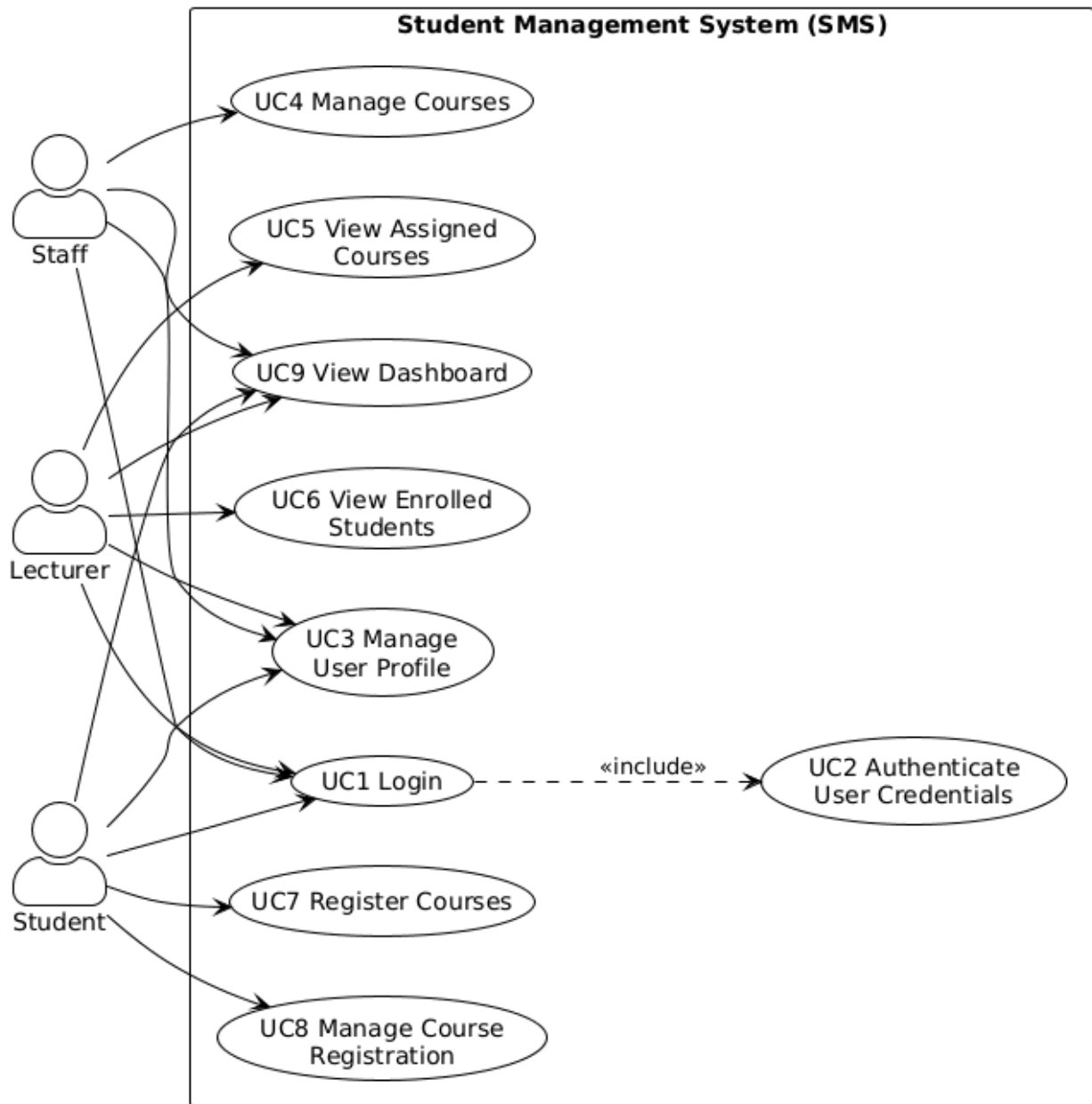


Table 1.1: Use Case - Login (UC1)

Aspect	Description
Actor	Staff, Lecturer, Student
Goal	Access the system with valid credentials
Precondition	User has a registered account in the system
Main Flow	1) User navigates to login page. 2) User enters Gmail and Password. 3) System validates credentials against database. 4) System checks user type (role). 5) System redirects to appropriate dashboard.
Alternative Flow	A1) Invalid credentials: display error message and prompt retry. A2) Inactive user: display access denied message.
Postcondition	User session is created; user is directed to role-specific dashboard.

Table 1.2: Use Case - Authenticate User Credentials (UC2)

Aspect	Description
Actor	System
Goal	Verify user identity for secure access
Precondition	User credentials are submitted via the login form
Main Flow	1) System receives username and password. 2) System queries user record from tb_user.

	3) System verifies password using BCrypt. 4) System validates user account status.
Alternative Flow	A1) Password mismatch: increment failed attempts. A2) User account locked: display account locked message.
Postcondition	User is authenticated and session token is generated.

Table 1.3: Use Case - Manage User Profile (UC3)

Aspect	Description
Actor	Staff, Lecturer, Student
Goal	Update personal information and account security
Precondition	User is logged in
Main Flow	1) User navigates to Profile page. 2) System displays current user information. 3) User edits details (Name, Email, Phone). 4) User optionally changes password. 5) System validates input and updates tb_user. 6) System displays success confirmation.
Alternative Flow	A1) Password mismatch: display error and prevent update. A2) Email already in use: display duplicate error. A3) Weak password: display requirements and reject.
Postcondition	User profile is updated in the database.

Table 1.4: Use Case - Manage Courses (UC4)

Aspect	Description
Actor	Staff
Goal	Create, modify, and remove course offerings
Precondition	Staff is logged in
Main Flow	Add Course: 1) Staff navigates to Course Management. 2) Staff fills course details (Code, Name, Section, Credits). 3) Staff selects lecturer from dropdown. 4) System validates course code uniqueness. 5) System inserts record into tb_course. 6) System displays success. Edit Course: 1) Staff selects course to edit. 2) Staff modifies course details. 3) System updates tb_course record. Delete Course: 1) Staff selects course to delete. 2) System checks whether the course has existing registrations. 3) If confirmed, system deletes the course record from tb_course. 4) System displays success message.
Alternative Flow	A1) Duplicate course code: display error. A2) Course has registrations: prompt confirmation or block deletion. A3) Invalid lecturer ID: display error.

Postcondition	Course catalog is updated.
---------------	----------------------------

Table 1.5: Use Case - View Assigned Courses (UC5)

Aspect	Description
Actor	Lecturer
Goal	View all courses assigned to the lecturer
Precondition	Lecturer is logged in
Main Flow	1) Lecturer navigates to Dashboard. 2) System queries tb_course WHERE u_id_lecturer = lecturer_id. 3) System displays course list (Code, Name, Section, Credits). 4) Lecturer can select course for more details.
Alternative Flow	A1) No courses assigned: display 'No courses assigned yet' message.
Postcondition	Course list is displayed with action buttons.

Table 1.6: Use Case - View Enrolled Students (UC6)

Aspect	Description
Actor	Lecturer
Goal	View all students enrolled in a specific course
Precondition	Lecturer is logged in and has selected a course

Main Flow	1) Lecturer clicks 'View Students'. 2) System queries tb_registration and tb_user. 3) System joins data to obtain student details. 4) System displays a paginated student list.
Alternative Flow	A1) No students enrolled: display 'No students enrolled' message.
Postcondition	Student enrollment list is displayed.

Table 1.7: Use Case - Register Courses (UC7)

Aspect	Description
Actor	Student
Goal	Search and enroll in available courses
Precondition	Student is logged in
Main Flow	1) Student navigates to Course Registration. 2) System displays all available courses. 3) Student searches by course code or name. 4) Student clicks 'View Sections & Register'. 5) System displays available sections for course. 6) System shows current enrollment and max capacity. 7) Student selects section and clicks 'Register'. 8) System checks max capacity. 9) System inserts registration record into tb_registration. 10) System displays success message.

Alternative Flow	A1) Section full: display 'Section is full' error. A2) Already registered: display 'Already enrolled' message. A3) Credits exceed limit: display warning.
Postcondition	Student is added to tb_registration with status 'Active'.

Table 1.8: Use Case - Manage Course Registration (UC8)

Aspect	Description
Actor	Student
Goal	Modify or remove course enrollments
Precondition	Student is logged in and has active registrations
Main Flow	View Enrollments: 1) Student navigates to My Courses. 2) System displays all registered courses. Change Section: 1) Student selects 'Change Section'. 2) System displays available sections. 3) Student confirms section change. 4) System updates tb_registration. Drop Course: 1) Student clicks 'Drop Course'. 2) System requests confirmation. 3) Student confirms. 4) System deletes registration record.
Alternative Flow	A1) Drop deadline passed: display restriction message.

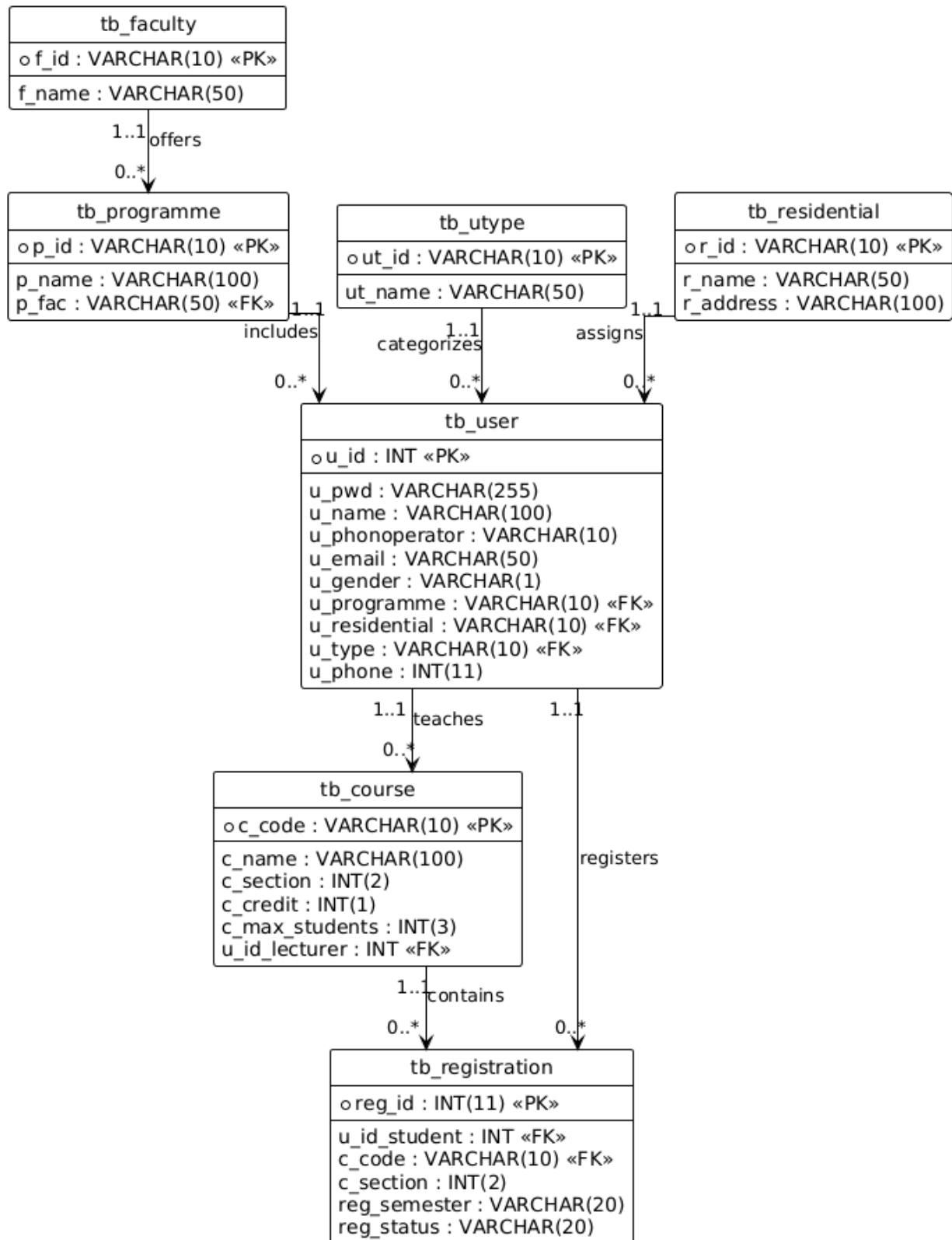
	A2) No other sections available: prevent section change.
Postcondition	Registration is modified or deleted as requested.

Table 1.9: Use Case - View Dashboard (UC9)

Aspect	Description
Actor	Staff, Lecturer, Student
Goal	View role-specific summary and important information
Precondition	User is logged in
Main Flow	Staff Dashboard: 1) System displays total courses, students, lecturers. 2) System shows pending registrations. 3) System provides quick links to management functions. Lecturer Dashboard: 1) System displays assigned courses. 2) System shows enrolled student count per course. Student Dashboard: 1) System displays student program and welcome message. 2) System provides shortcuts to course registration and my courses. 3) System shows current registration status.
Alternative Flow	None

Postcondition	Dashboard is displayed with relevant information.
---------------	---

Figure 1.1: Entity Relationship Diagram (ERD)



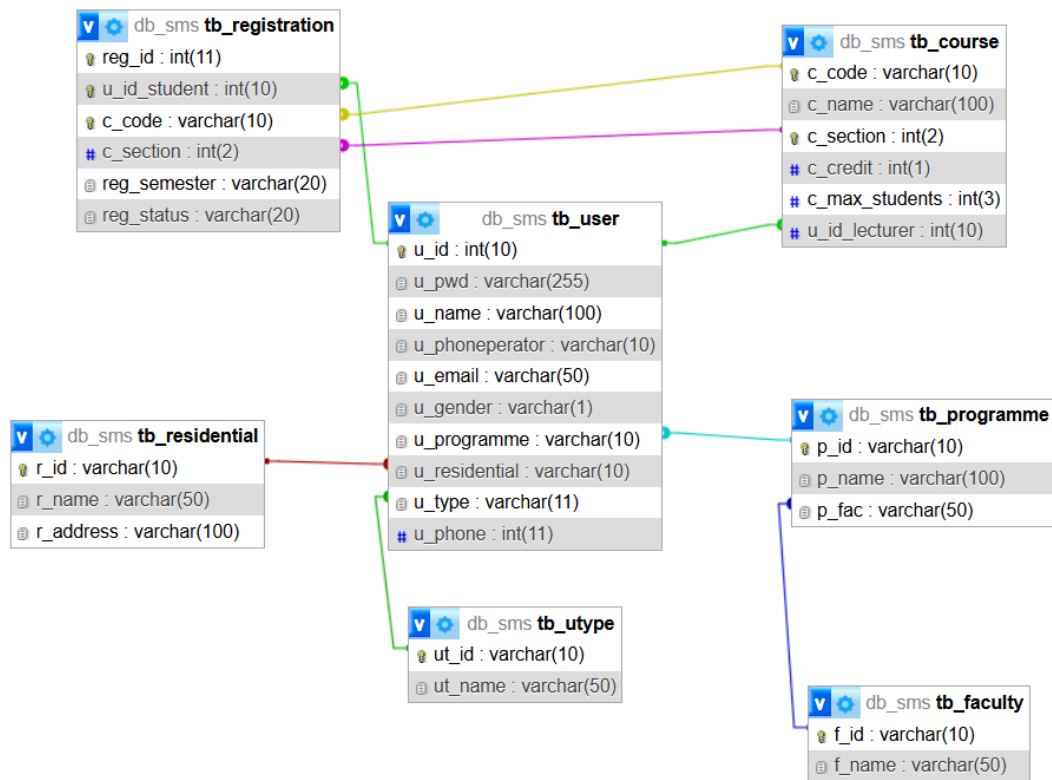
1.3 Development Software, Language, Technology, and Tools

Table 1.10: Development Software, Language, Technology, and Tools

Category	Technology/ Tool	Version	Purpose	Official URL
Server-Side Language	PHP	8.2.12	Core backend logic and processing	https://www.php.net/
Database Management System	MariaDB (MySQL)	10.4.32	Storing and retrieving system data	https://mariadb.org/
Web Server	Apache HTTP Server	2.4.x	Serving web pages on localhost	https://httpd.apache.org/
Frontend Framework	Bootstrap	5.3.3	Responsive UI design and styling	https://getbootstrap.com/
Local Server Environment	XAMPP	8.2.12	Local developme nt stack (Apache, MySQL, PHP)	https://www.apachefriends.org/

Database Administration	phpMyAdmin	5.2.1	GUI tool for managing MySQL databases	https://www.phpmyadmin.net/
Integrated Development Environment	Visual Studio Code	Latest	Code editing and debugging	https://code.visualstudio.com/
Web Browser	Google Chrome	Latest	Testing and running the application	https://www.google.com/chrome/

1.4 Database Design (phpMyAdmin screenshot)



1.5 Detail development steps with screenshots

1.5.1 Student Function

a) Register Courses

 student_course_register.php:1 - Search and register for courses:

```
<?php
// Get unique courses with search
$sql = "SELECT c_code, c_name, c_credit, COUNT(*) as section_count
        FROM tb_course";
if (!empty($search)) {
    $sql .= " WHERE c_code LIKE '%" . mysqli_real_escape_string($con, $search) . "%'
            OR c_name LIKE '%" . mysqli_real_escape_string($con, $search) . "%'
        }
    }
}
```

b) View Current & Previous Semester Courses

 student_courses_view.php:1 - Display registered courses with drop functionality

```
// Handle Drop
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) && $_POST['action'] === 'drop') {
    $token = $_POST['csrf_token'] ?? '';
    if (!csrf_verify($token)) {
        $msg = 'CSRF verification failed.';
    } else {
        $reg_id = intval($_POST['reg_id'] ?? 0);
        if ($reg_id > 0) {
            $delSql = "DELETE FROM tb_registration WHERE reg_id = ? AND u_id_student = ?";
            $stmt = mysqli_stmt_init($con);
            if (mysqli_stmt_prepare($stmt, $delSql)) {
                mysqli_stmt_bind_param($stmt, "ii", $reg_id, $u_id);
                if (mysqli_stmt_execute($stmt)) {
                    $msg = "Course dropped successfully.";
                } else {
                    $msg = "Error dropping course.";
                }
            }
            mysqli_stmt_close($stmt);
        }
    }
}

// Handle Change Section
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) && $_POST['action'] === 'change_section') {
    $token = $_POST['csrf_token'] ?? '';
    if (!csrf_verify($token)) {
        $msg = 'CSRF verification failed.';
    } else {
        $reg_id = intval($_POST['reg_id'] ?? 0);
        $new_section = intval($_POST['new_section'] ?? 0);
    }
}
```


c) Search Course

Built into  student_course_register.php:58-60 :

```
<?php
<form action="" method="GET" class="d-flex gap-2">
    <input type="text" name="search" class="form-control"
        placeholder="🔍 Search by Course Code or Name..."
        value="<?php echo htmlspecialchars($search); ?>">
```

d) Modify Registration

 student_courses_view.php:44-53 - Change sections:

```
<?php
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) && $_POST['action']
```

e) Cancel Registration

 student_courses_view.php:20-39 - Drop course:

```
<?php
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) && $_POST['action']
    $delSql = "DELETE FROM tb_registration WHERE reg_id = ? AND u_id_student = ?";
```

f) Edit Profile

 student_profile.php:1 &  student_profile_process.php - Update student information

```
28
29 // Password update logic
30 if (!empty($new_pass)) {
31     if ($new_pass !== $conf_pass) {
32         header('Location: student_profile.php?msg=Passwords do not match');
33         exit();
34     }
35     $hashed_pwd = password_hash($new_pass, PASSWORD_DEFAULT);
36     $sql = "UPDATE tb_user SET u_name=?, u_email=?, u_phone=?, u_pwd=? WHERE u_id=?";
37
38     $stmt = mysqli_stmt_init($con);
39     if (mysqli_stmt_prepare($stmt, $sql)) {
40         mysqli_stmt_bind_param($stmt, "ssisi", $u_name, $u_email, $u_phone, $hashed_pwd, $u_id);
41     } else {
42         header('Location: student_profile.php?msg=SQL Error');
43         exit();
44     }
45 }
```

1.5.2 Lecturer Function

a) View Assigned Courses for Each Semester

🔗 lecturer.php:24-45 - Display courses assigned to lecturer:

```
<?php
$sql = "SELECT c_code, c_name, c_section, c_credit, c_max_students
        FROM tb_course
        WHERE u_id_lecturer = ?
        ORDER BY c_code ASC";
```

b) View Student List for Each Course

🔗 lecturer_view_students.php:1 - Search and filter students in course:

```
<?php
$whereClause = "WHERE r.c_code = ? AND r.c_section = ?";
if (!empty($search)) {
    $whereClause .= " AND (u.u_name LIKE ? OR u.u_email LIKE ?)";
}
```

c) View Course Details

🔗 lecturer_course_details.php - Display course information

d) View Student Details

🔗 lecturer_view_student_detail.php - View individual student information

```

lecturer_course_details.php X
lecturer_course_details.php
1  <?php
2  include 'dbconnect.php';
3  include 'headerlec.php';
4
5  // Security: Ensure User is Lecturer
6  if (!isset($_SESSION['u_type']) || $_SESSION['u_type'] != '02') {
7      if (isset($_SESSION['u_type']) && $_SESSION['u_type'] == '01') {
8          header("Location: staff.php");
9      } else {
10         header("Location: login.php");
11     }
12     exit();
13 }
14
15 // Validate input
16 if (!isset($_GET['c_code']) || !isset($_GET['c_section'])) {
17     echo '<div class="container mt-5"><div class="alert alert-danger">Missing course code or section.</div></div>';
18     include 'footer.php';
19     exit();
20 }
21
22 $c_code = $_GET['c_code'];
23 $c_section = $_GET['c_section'];
24 $lecturer_id = $_SESSION['u_id'];
25
26 // Fetch course and ensure it belongs to this lecturer
27 $sql = "SELECT c_code, c_name, c_section, c_credit, c_max_students, u_id_lecturer FROM tb_course WHERE c_code = ? AND c_section = ?";
28 $stmt = mysqli_stmt_init($con);
29 if (!mysqli_stmt_prepare($stmt, $sql)) {
30     echo '<div class="container mt-5"><div class="alert alert-danger">Unable to load course details.</div></div>';
31     include 'footer.php';
32     exit();
33 }
34
35 // Pagination
36 $perPage = 15;
37 $currentPage = intval($_GET['page'] ?? 1);
38 $offset = ($currentPage - 1) * $perPage;
39 $totalPages = ceil($totalRows / $perPage);
40
41 // Get students with pagination
42 $sql = "SELECT r.reg_id, u.u_id, u.u_name, u.u_email, u.u_phone, u.u_phoneoperator,
43         u.u_programme, r.reg_status
44        FROM tb_registration r
45        JOIN tb_user u ON r.u_id_student = u.u_id
46        $whereClause
47        ORDER BY u.u_name ASC
48        LIMIT ? OFFSET ?";
49
50 $stmt = mysqli_stmt_init($con);
51 mysqli_stmt_prepare($stmt, $sql);
52 $params[] = $perPage;
53 $params[] = $offset;
54 $types .= "ii";
55 mysqli_stmt_bind_param($stmt, $types, ...$params);
56 mysqli_stmt_execute($stmt);
57 $result = mysqli_stmt_get_result($stmt);

```

1.5.3 Staff Function

a) Add New Course with Max Students

staff_course_add.php:25-60 :

```
<?php
<input type="text" name="c_code" placeholder="e.g. SECP3723" required>
<input type="number" name="c_max_students" value="30" min="1" required>
<select name="u_id_lecturer"><!-- Assign Lecturer --></select>
```

b) Modify Course Details

staff_course_edit.php & staff_course_modify_process.php

c) Delete Course

staff_course_delete_process.php

d) Amend/Cancel Registration

staff.php:120 - Dashboard with registration management

e) Dashboard

staff.php:48-85 - Statistics and quick actions:

```
// ===== UPDATE COURSE =====
if ($action === 'edit') {
    // New Data (SET clause)
    $c_code = strtoupper(htmlspecialchars(trim($_POST['c_code'] ?? '')));
    $c_section = htmlspecialchars($_POST['c_section'] ?? '');
    $c_name = htmlspecialchars(trim($_POST['c_name'] ?? ''));
    $c_credit = intval($_POST['c_credit'] ?? 0);
    $c_max = intval($_POST['c_max_students'] ?? 0);
    $u_id_lecturer = !empty($_POST['u_id_lecturer']) ? intval($_POST['u_id_lecturer']) : NULL;

    $errors = [];
    if (empty($c_code)) $errors[] = "Course code is required";
    if (empty($c_name)) $errors[] = "Course name is required";
    if ($c_credit <= 0) $errors[] = "Credits must be greater than 0";
    if ($c_max <= 0) $errors[] = "Max students must be greater than 0";

    if (!empty($errors)) {
        $_SESSION['errors'] = $errors;
        header("Location: staff_course_edit.php?code=$old_code&sec=$old_section");
        exit();
    }

    try {
        $sql = "UPDATE tb_course
                SET c_code=?, c_section=?, c_name=?, c_credit=?, c_max_students=?, u_id_lecturer=?
                WHERE c_code=? AND c_section=?";

        $stmt = mysqli_stmt_init($con);
        if (!mysqli_stmt_prepare($stmt, $sql)) {
```

```

42
43 // SECOND: Delete the course
44 $sql = "DELETE FROM tb_course WHERE c_code = ? AND c_section = ?";
45 $stmt = mysqli_stmt_init($con);
46
47 if (mysqli_stmt_prepare($stmt, $sql)) {
48     mysqli_stmt_bind_param($stmt, "ss", $c_code, $c_section);
49
50     // 4. Execute with Error Handling
51     if (mysqli_stmt_execute($stmt)) {
52         // Send notification to affected students
53         foreach ($affected_students as $student) {
54             $subject = "Course Deleted: $c_code";
55             $msg = "Dear " . $student['u_name'] . ",\n\nThe course $c_code (Section $c_section) has been deleted from the system.\n\nYo
56             send_notification_email($student['u_email'], $subject, $msg);
57         }
58
59         header("Location: staff.php?msg=" . urlencode('Course Deleted Successfully'));
60         exit();
61     } else {
62         $errno = mysqli_errno($con);
63         if ($errno == 1451) {
64             $error_msg = "Cannot delete course. Students are currently registered for it.";
65         } else {
66             $error_msg = "Database Error: " . mysqli_error($con);
67         }
68         header("Location: staff.php?msg=" . urlencode($error_msg));
69         exit();
70     }
71 } else {

```

e) Dashboard

staff.php:48-85 - Statistics and quick actions:

```

<?php
$total_courses = mysqli_fetch_assoc(mysqli_query($con, "SELECT COUNT(*) as c FROM tb_c
$total_registrations = mysqli_fetch_assoc(mysqli_query($con, "SELECT COUNT(*) as c FRC
$pending_registrations = mysqli_fetch_assoc(mysqli_query($con, "SELECT COUNT(*) as c F

```

1.5.4 Compulsory and Additional Function

a) Encrypt/Hash Password

registerprocess.php:84 :

```

<?php
$hashed_password = password_hash($fpwd, PASSWORD_BCRYPT, ['cost' => 12]);

```

loginprocess.php:23-30 - Verify hashed password:

```

<?php
$pwdCheck = password_verify($fpwd, $row['u_pwd']);
if ($pwdCheck == false) {
    header('Location: login.php?error=wrongpwd');
}

```

b) Notification, Confirmation, Verification

📄 email_helper.php:37-46 - Send email notifications via SMTP:

```
<?php
function send_notification_email($to, $subject, $message) {
    $logDate = date('Y-m-d H:i:s');
    if (send_via_smtp($to, $subject, $message)) {
        file_put_contents('email_log.txt', "[$logDate] Status: ✓ Sent via SMTP" . PHP_EOL);
        return true;
    }
}
```

c) SQL Injection Prevention

📄 loginprocess.php:11-17 - Prepared statements:

```
<?php
$sql = "SELECT u_id, u_pwd, u_name, u_type FROM tb_user WHERE u_email = ?";
$stmt = mysqli_stmt_init($con);
mysqli_stmt_prepare($stmt, $sql);
mysqli_stmt_bind_param($stmt, "s", $femail);
mysqli_stmt_execute($stmt);
```

d) Password Double Field Check

registerprocess.php:47-49 :

```
<?php
if ($fpwd !== $fpwd_confirm) {
    $errors[] = "Passwords do not match";
}
```

e) Password Strength Validation

registerprocess.php:51-54 - Enforces password complexity

f) Fix Login Error - Proper Role-Based Redirection

loginprocess.php:40-54 :

```
<?php
if ($row['u_type'] == '01') {
    header('Location: staff.php'); // Staff dashboard
} elseif ($row['u_type'] == '02') {
    header('Location: lecturer.php'); // Lecturer dashboard
} elseif ($row['u_type'] == '03') {
    header('Location: student.php'); // Student dashboard
}
```

g) Auto-Approved Registration

Configured in registration process when capacity allows

h) CSRF Protection

csrf.php:1 :

```
<?php
function csrf_token() {
    if (empty($_SESSION['csrf_token'])) {
        $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
    }
    return $_SESSION['csrf_token'];
}
```

1.5.5 Email Function and storing Email Credentials Securely (.env)

🐞 email_helper.php

```
13 function format_text_to_html($text) {
23     // Convert # Heading to <h2>
24     $text = preg_replace('/^# (.*)$/m', '<h2>$1</h2>', $text);
25
26     // Convert ## Heading to <h3>
27     $text = preg_replace('/^## (.*)$/m', '<h3>$1</h3>', $text);
28
29     // Convert line breaks to <br>
30     $text = nl2br($text);
31
32     // Wrap in HTML body tags
33     $html = '<html><head><meta charset="UTF-8"></head><body style="font-family
34
35     return $html;
36 }
37
38 function send_notification_email($to, $subject, $message) {
39     $logDate = date('Y-m-d H:i:s');
40     $logMessage = "[ $logDate ] To: $to | Subject: $subject | Message: " . substr($message, 0, 100);
41     file_put_contents('email_log.txt', $logMessage, FILE_APPEND);
42
43     if (send_via_smtp($to, $subject, $message)) {
44         file_put_contents('email_log.txt', "[ $logDate ] Status: / Sent via SMTP", FILE_APPEND);
45     }
46 }
```



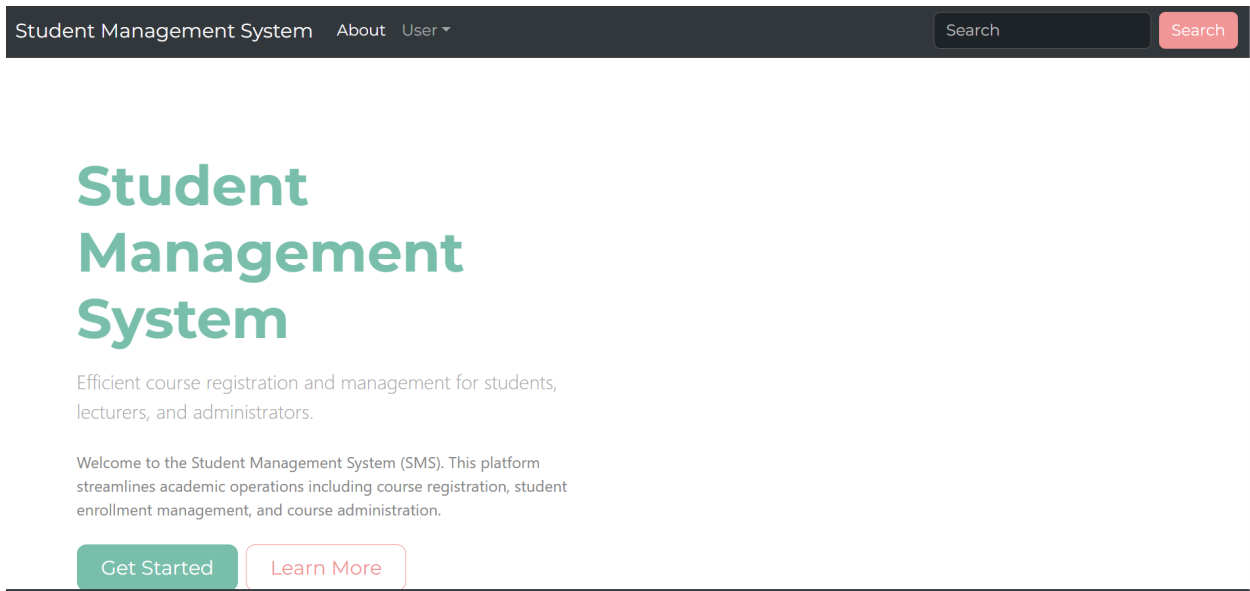
```

● PS C:\xampp\htdocs\sms> composer require vlucas/phpdotenv
./composer.json has been created
Running composer update vlucas/phpdotenv
Loading composer repositories with package information
Updating dependencies
Lock file operations: 6 installs, 0 updates, 0 removals
  - Locking graham-campbell/result-type (v1.1.4)
  - Locking phpoption/phpooption (1.9.5)
  - Locking symfony/polyfill-ctype (v1.33.0)
  - Locking symfony/polyfill-mbstring (v1.33.0)
  - Locking symfony/polyfill-php80 (v1.33.0)
  - Locking vlucas/phpdotenv (v5.6.3)
Writing lock file
Installing dependencies from lock file (including require-dev)
Package operations: 6 installs, 0 updates, 0 removals
  - Downloading phpooption/phpooption (1.9.5)
  - Downloading graham-campbell/result-type (v1.1.4)
  - Downloading vlucas/phpdotenv (v5.6.3)
  - Installing symfony/polyfill-php80 (v1.33.0): Extracting archive
  - Installing symfony/polyfill-mbstring (v1.33.0): Extracting archive
  - Installing symfony/polyfill-ctype (v1.33.0): Extracting archive
  - Installing phpooption/phpooption (1.9.5): Extracting archive
  - Installing graham-campbell/result-type (v1.1.4): Extracting archive
  - Installing vlucas/phpdotenv (v5.6.3): Extracting archive
Generating autoload files
6 packages you are using are looking for funding.
Use the `composer fund` command to find out more!
No security vulnerability advisories found.
Using version ^5.6 for vlucas/phpdotenv

```

1.6 System Interface for all users

Figure 1.6.1 Home page for all user



The screenshot shows the home page of the Student Management System. At the top is a dark navigation bar with the text "Student Management System" on the left, "About" and "User" with a dropdown arrow in the center, and two "Search" buttons on the right. The main content area has a large green heading "Student Management System". Below it, a paragraph describes the system as efficient for students, lecturers, and administrators. Another paragraph welcomes users and lists features like course registration, student enrollment management, and course administration. At the bottom of the main content area are two buttons: "Get Started" in green and "Learn More" in red.

Student Management System About User ▾ Search Search

Student Management System

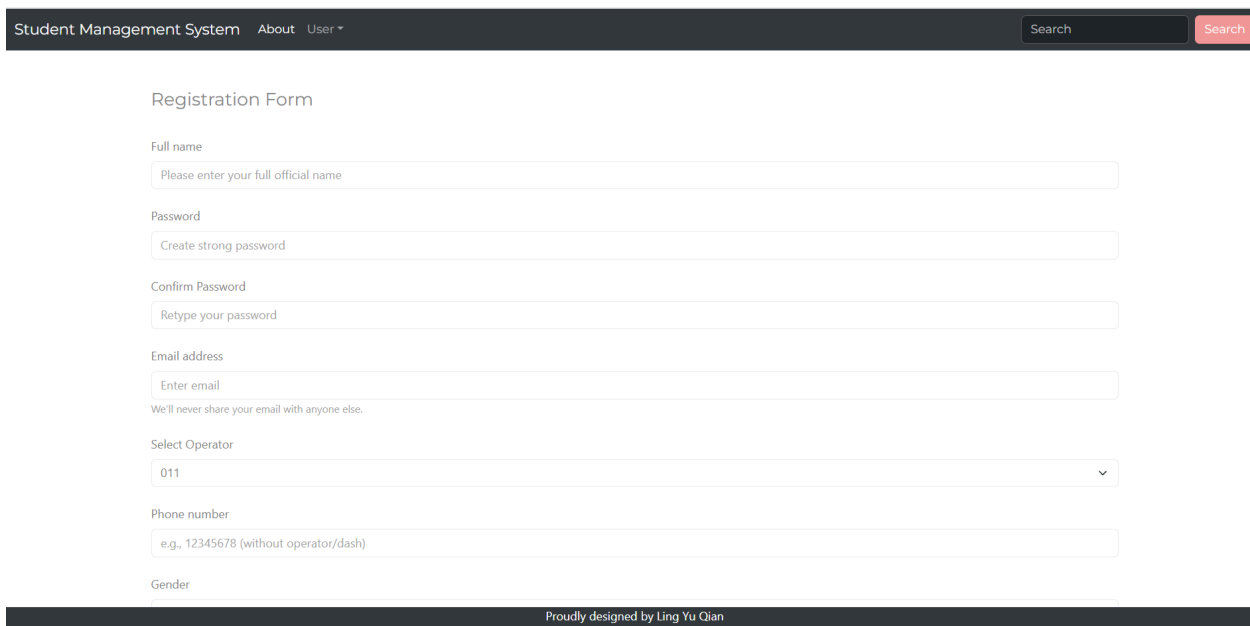
Efficient course registration and management for students, lecturers, and administrators.

Welcome to the Student Management System (SMS). This platform streamlines academic operations including course registration, student enrollment management, and course administration.

Get Started Learn More

Figure 1.3: Storing Email Credentials Securely (.env)

Figure 1.6.2 Home page for student register



The screenshot shows the registration form of the Student Management System. It has the same dark navigation bar as Figure 1.6.1. The main content area is titled "Registration Form" and contains several input fields: "Full name" with a placeholder "Please enter your full official name", "Password" with a placeholder "Create strong password", "Confirm Password" with a placeholder "Retype your password", "Email address" with a placeholder "Enter email" and a note "We'll never share your email with anyone else.", "Select Operator" with a dropdown menu showing "011", "Phone number" with a placeholder "e.g., 12345678 (without operator/dash)", and "Gender". At the bottom of the page is a dark footer with the text "Proudly designed by Ling Yu Qian".

Student Management System About User ▾ Search Search

Registration Form

Full name
Please enter your full official name

Password
Create strong password

Confirm Password
Retype your password

Email address
Enter email
We'll never share your email with anyone else.

Select Operator
011 ▾

Phone number
e.g., 12345678 (without operator/dash)

Gender

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Figure 1.6.2.1 Criteria for student to register

Registration Conditions:

1. **Full name** – Required (not empty)
2. **Email** – Required, must be valid email format, and cannot be already registered
3. **Password** – Required and must match complexity requirements:
 - At least 8 characters long
 - At least one uppercase letter
 - At least one lowercase letter
 - At least one number
 - At least one special character (!@#\$%^&*...)
4. **Confirm Password** – Must match the password field exactly
5. **Gender** – Required (Male or Female)
6. **Programme** – Required (from dropdown)
7. **Operator** – Optional dropdown (011, 012, 013, 017, 019)
8. **Phone number** – Required (8 digits, without operator/dash)
9. **Residential College** – Required (from dropdown)

Any validation failure shows error messages and returns user to the form.

Figure 1.6.3 Student receive email after successfully register

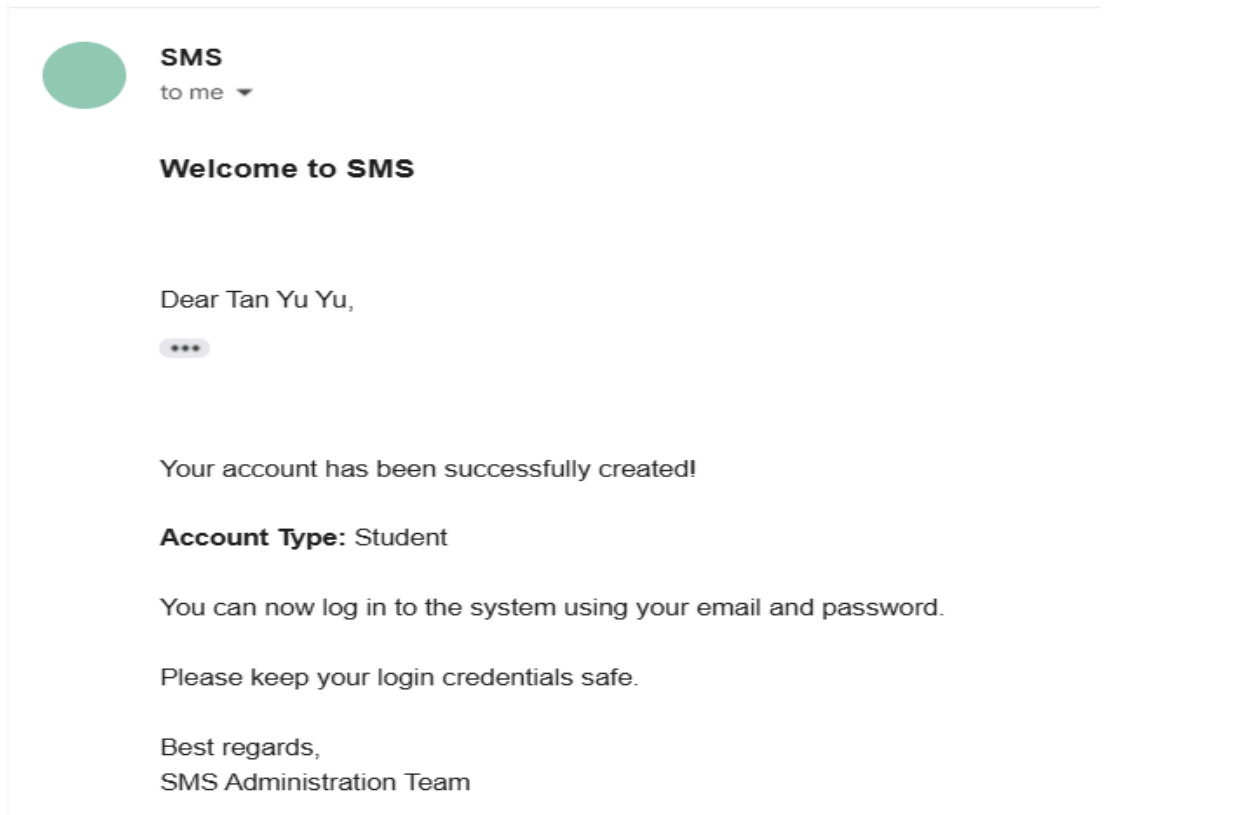


Figure 1.6.4 Home page after student login

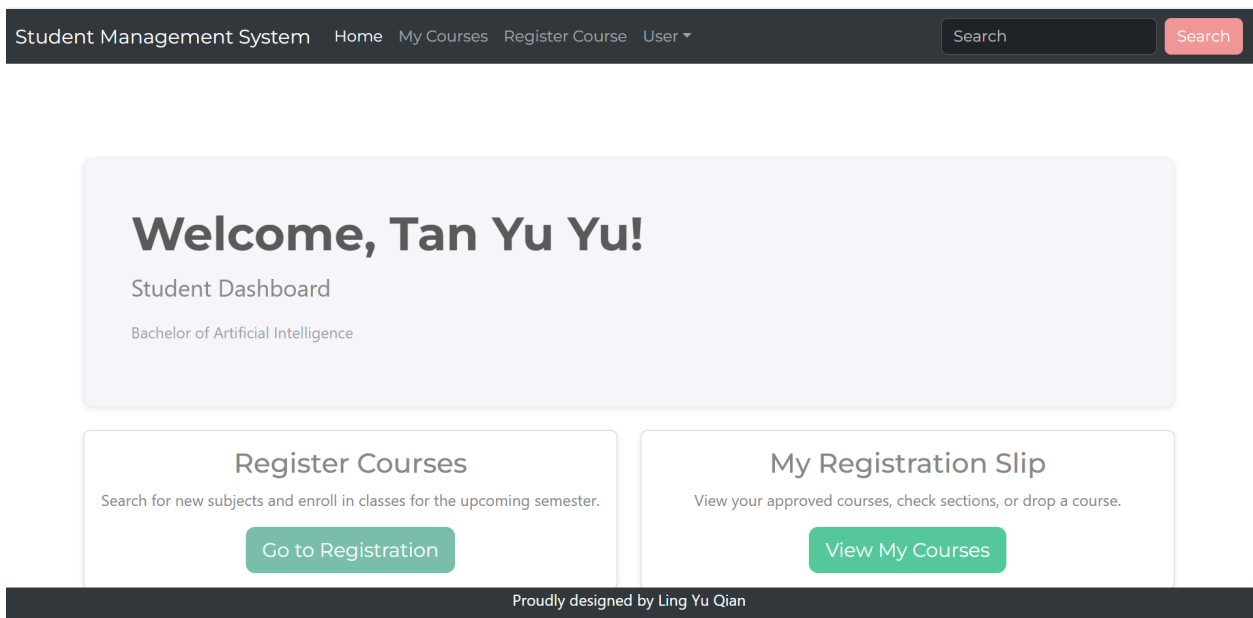


Figure 1.6.5 Student Register Courses

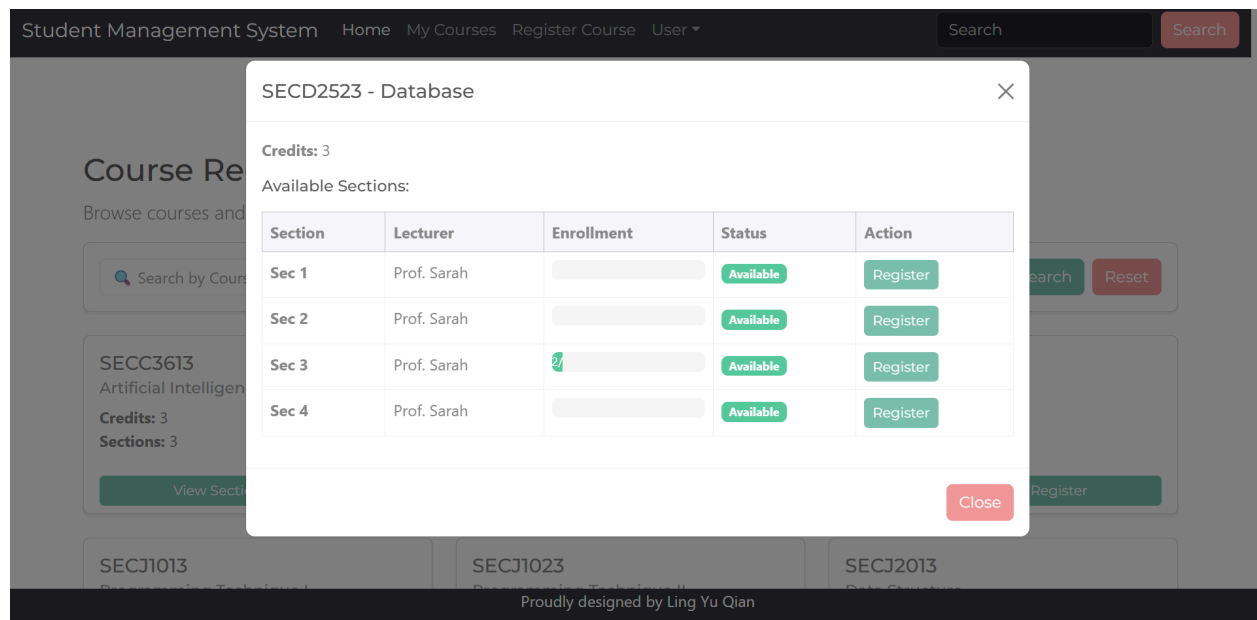


Figure 1.6.6 Student receive email notification

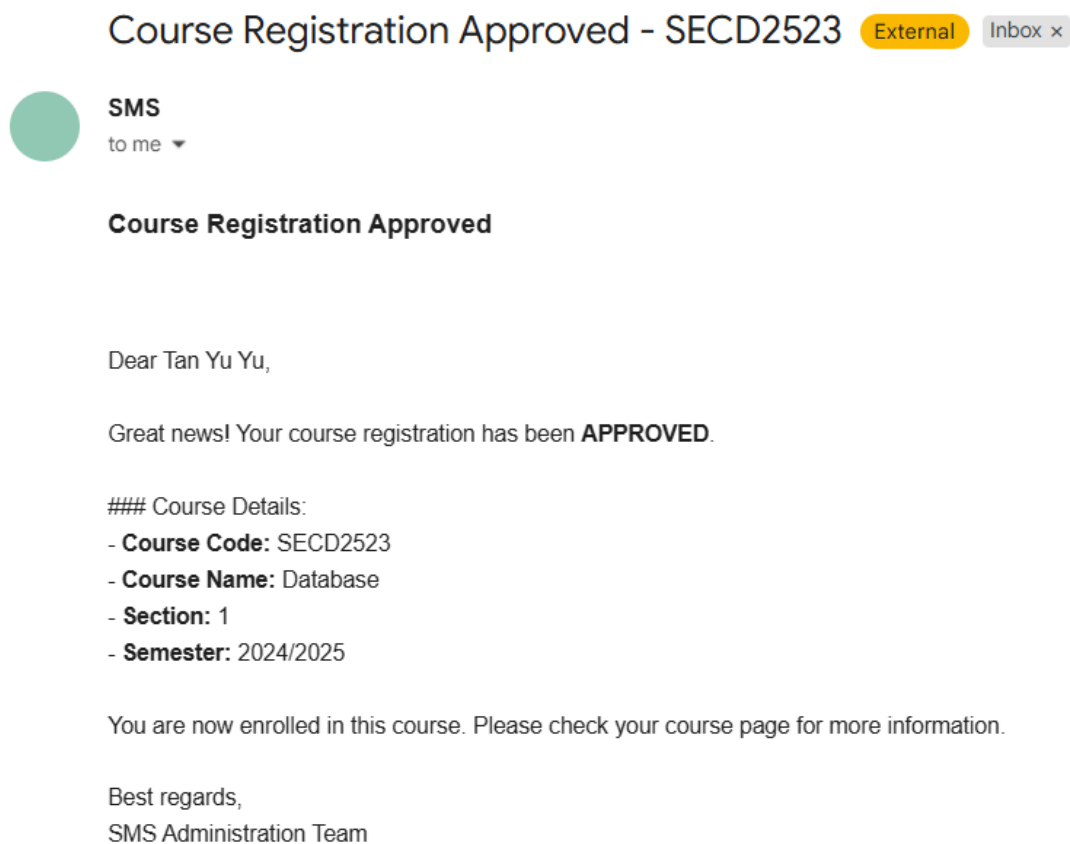


Figure 1.6.7 Student View current semester

Student Management System

Home

My Courses

Register Course

User

Search

Search

My Courses

Registration Successful

Current Semester (2024/2025)

Previous Semesters

Semester	Code	Course Name	Section	Credit	Status	Action
2024/2025	SECD2523	Database	1	3	Approved	Switch section... Switch Drop
2024/2025	SECI1013	Discrete Structure	1	3	Approved	Switch section... Switch Drop
2024/2025	SECJ1023	Programming Technique II	1	3	Approved	Switch section... Switch Drop
2024/2025	SECJ2013	Data Structure	1	3	Approved	Switch section... Switch Drop

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Figure 1.6.7.1 Student View previous semester

Student Management System

Home

My Courses

Register Course

User

Search

Search

My Courses

Current Semester (2024/2025)

Previous Semesters

Semester	Code	Course Name	Section	Credit	Status
2023/2024	SECI1013	Discrete Structure	1	3	Approved
2023/2024	SECJ1013	Programming Technique I	1	3	Approved
2023/2024	UHAS1172	Dinamika Malaysia	1	2	Approved
2023/2024	UKQF2082	Co-Curriculum	1	2	Approved
Total Credits:				10	

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Figure 1.6.8 Student Search course

Student Management System
Home
My Courses
Register Course
User
Search
Search

Course Registration

Browse courses and view available sections

Search
Reset

SECJ1013
Programming Technique I
Credits: 3
Sections: 4
View Sections & Register

SECJ1023
Programming Technique II
Credits: 3
Sections: 4
View Sections & Register

SECJ2013
Data Structure
Credits: 3
Sections: 4
View Sections & Register

SECJ3553
Enrolled

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Figure 1.6.9 Student Modify registration

Student Management System
Home
My Courses
Register Course
User
Search
Search

My Courses

Section changed to SECD2523 (Section 2) successfully.

Current Semester (2024/2025)
Previous Semesters

Semester	Code	Course Name	Section	Credit	Status	Action
2024/2025	SECD2523	Database	2	3	Approved	Switch section... Switch Drop
2024/2025	SECJ1013	Discrete Structure	1	3	Approved	Switch section... Switch Drop
2024/2025	SECJ1023	Programming Technique II	1	3	Approved	Switch section... Switch Drop
2024/2025	SECJ2013	Data Structure	1	3	Approved	Switch section... Switch Drop

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Figure 1.6.10 Student Cancel/drop registration

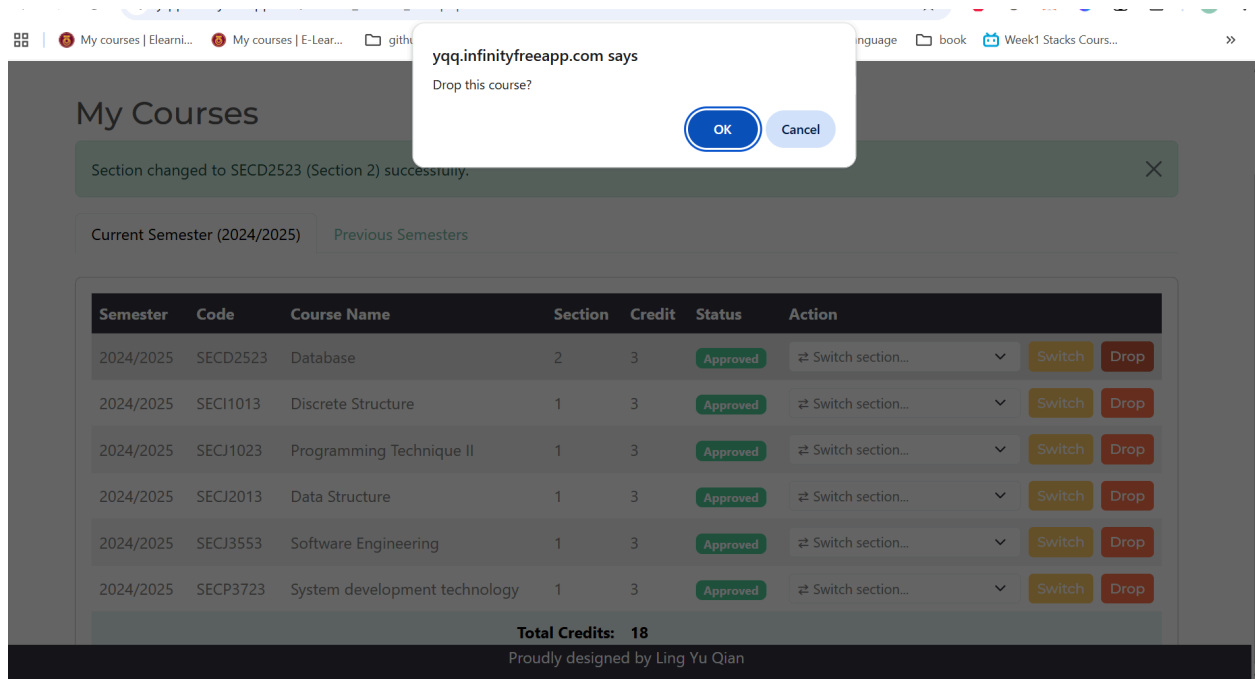


Figure 1.6.10.1 result of Student Cancel/drop registration

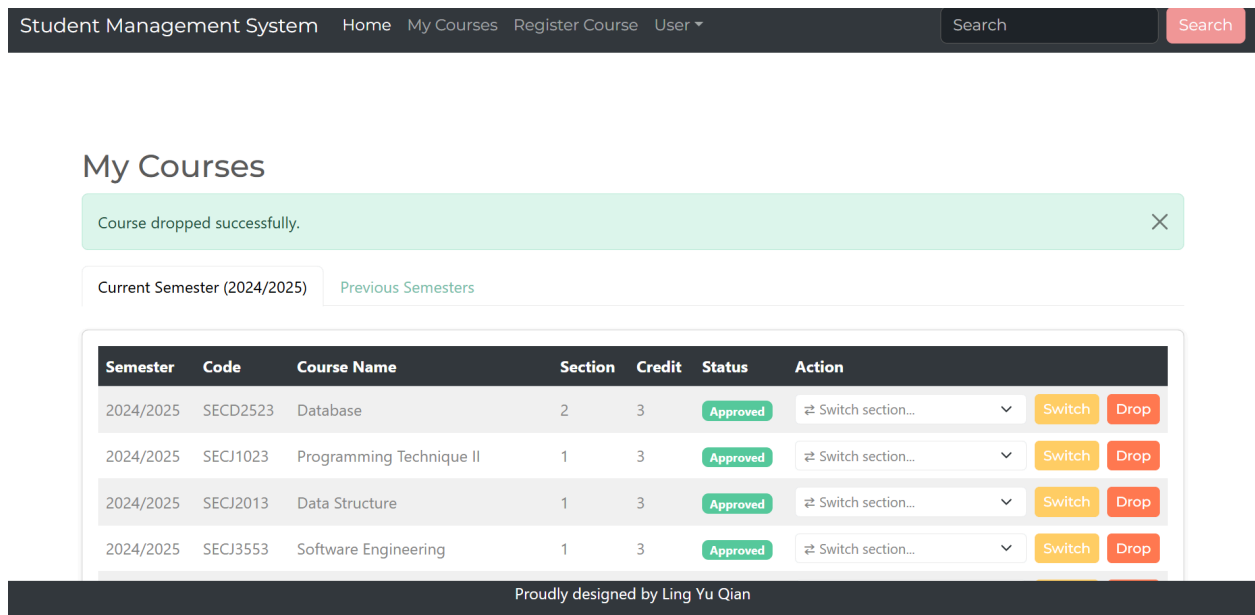


Figure 1.6.11 Student Edit profile

Edit Profile

Full Name

Tan Yu Yu

Email

yuyu@gmail.com

Phone Number

34567890

Change Password (leave blank to keep current)

New Password

Confirm New Password

Confirm New Password

Update Profile

Cancel

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Figure 1.6.12 Lecturer View assigned courses for each semester

Student Management System

Dashboard

Student

Academic

User

Search

Search

Lecturer Dashboard

Code	Name	Section	Credits	Max Students	Actions
SECI1013	Discrete Structure	1	3	31	View Details View Students
SECI1013	Discrete Structure	3	3	31	View Details View Students
SECI1013	Discrete Structure	4	3	31	View Details View Students
SECJ2013	Data Structure	1	3	35	View Details View Students
SECJ2013	Data Structure	2	3	35	View Details View Students
SECJ2013	Data Structure	3	3	35	View Details View Students
SECJ2013	Data Structure	4	3	35	View Details View Students
SECP2013	Data Engineering	1	3	30	View Details View Students

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Figure 1.6.13 Lecturer View student list for each course

Student Management System
Dashboard
Student
Academic
User
Search
Search

Dashboard / SECI1013

Student List

Section 1

SECI1013 - Discrete Structure

#	Student Details	Contact Info	Programme	Action
1	Tan Yu Qian	Email: yqling29@gmail.com Phone: 01155934038	FAICH	View Details
2	Test	Email: test@gmail.com Phone: 011557643521	SECPH	View Details

Back to Dashboard

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Figure 1.6.14 Lecturer View course details

Student Management System
Dashboard
Student
Academic
User
Search
Search

Course Details: SECI1013 - Discrete Structure

Code	SECI1013
Name	Discrete Structure
Section	1
Credit	3
Max Students	31
Current Registered	2

View Students Back to Dashboard

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Figure 1.6.15 Lecturer View student details

Student Management System

Dashboard

Student

Academic

User

Search

Search

Student Details

Student ID	14
Name	Tan Yu Qian
Email	yqling29@gmail.com
Phone	01155934038
Gender	F
Programme	Bachelor of Artificial Intelligence

Back to Student List

Dashboard

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Figure 1.6.16 Admin Add new course details with number of maximum students per course

Student Management System

Dashboard

Registrations

User

Search

Search

Add New Course

Course Code

SECP3133

Section

5

Course Name

High performance data processing

Credit Hours

3

Max Students

30

Assign Lecturer

Dr. Tan

Cancel

Save Course

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Figure 1.6.17 Admin Modify course details

Student Management System
Dashboard
Registrations
User
Search
Search

Edit Course Details

Course Code

SECC3613

Section

2

Course Name

Artificial Intelligence

Credit

3

Max Students

30

Assign Lecturer

Prof. Sarah

Update Course

Cancel

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Figure 1.6.18 Admin Delete course

My courses | Elearni...
My courses | E-Lear...
github
language
book
Week1 Stacks Cours...

Total Courses

Add Course

Create a new course or section for the next semester.

Add Course

Total Students

Registrations

Approve or reject pending student registrations.

View Registrations

Course Management

Filters

Course Code	Section	Course Name	Credit	Lecturer	Enrollment	Actions
SECC3613	2	Artificial Intelligence	3	Prof. Sarah	1/30	Edit Delete
SECC3613	3	Artificial Intelligence	3	Prof. Sarah	1/30	Edit Delete
SECC3613	4	Artificial Intelligence	3	Prof. Sarah	0/30	Edit Delete

yqq.infinityfreeapp.com says

Are you sure you want to delete this course? All registrations will also be deleted.

OK

Cancel

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https://yqq.infinityfreeapp.com/staff_course_delete_process.php?code=SECC3613&sec=2

Figure 1.6.19 Admin Amend or cancel registration

36

Student Management System
Dashboard
Registrations
User
Search
Search

Manage Student Registrations

Reg ID	Student Name	Course	Section	Status	Action
52	Tan Yu Yu	SECP3723	1	Approved	Amend Cancel
51	Tan Yu Yu	SECJ3553	1	Approved	Amend Cancel
50	Tan Yu Yu	SECJ2013	1	Approved	Amend Cancel
49	Tan Yu Yu	SECJ1023	1	Approved	Amend Cancel
47	Tan Yu Yu	SECD2523	2	Approved	Amend Cancel
46	Ling Yu An	SECC3613	3	Approved	Amend Cancel
45	Tan Yu Qian	SECI1013	1	Approved	Amend Cancel

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Figure 1.6.20 Dashboard

Welcome, Staff!

Staff Management Dashboard

Manage courses, registrations, and system administration

99
Total Courses

32
Total Registrations

0
Pending Approvals

5
Total Students

Add Course

Create a new course or section for the next semester.

Add Course

Manage Courses

Edit, delete, or view details of existing courses.

View Courses

Registrations

Approve or reject pending student registrations.

View Registrations

Course Management

Filter Courses

Course Code
Lecturer
Capacity Status

e.g., SECI
All Lecturers
All Statuses

Apply
Clear

Course Code	Section	Course Name	Credit	Lecturer	Enrollment	Actions
-------------	---------	-------------	--------	----------	------------	---------

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Figure 1.6.21 Email Notification

subject: Course Registration Approved - SECC3613
mailed-by: gmail.com
Signed by: gmail.com
security:  Standard encryption (TLS) [Learn more](#)
 : Important mainly because it was sent directly to you.

1.7 Localhost Setup

```
<?php
//DB parameters
$Servername="localhost";
$username="root";
$password="";
$dbname="db_sms";
//create connection
$con=mysqli_connect($Servername,$username,$password,$dbname);
//verify connection

?>
```

1.8 Hosting URL

<https://yqq.infinityfreeapp.com/>

1.9 User credentials

User Type	Gmail	Password
Student	ali@utm.my	12345678
Lecturer	yuyu@gmail.com	12345678

Staff	staff@utm.my	12345678
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